

Next, the team should have a shared document repository. Preferably, it should have a version control system. There are various version control software options. Some are proprietary software. Under open source, there are options like Subversion. Concurrent versioning systems (CVS), Git, etc. Even the most common office software products like Microsoft and Open Office suits have options to document and monitor changes. The team should adopt any of these options and maintain a change history.

Under concurrent engineering development processes, many changes will happen in parallel. Some of these changes will have inter-relations. When we make any change in the design or processes its impact on the other related areas needs to be evaluated, and the agreed changes incorporated into the documents.

It is a good practice to mention the related steps in each document. Establishing a lookup table with the document dependencies and the latest version numbers is also a good idea. Such documentation helps to ensure proper analysis of changes and helps to avoid unnecessary work of checking change impact for unrelated areas.

Development stages

We shall look into the stages of the Vee developmental model with the example of the charge controller. So far we have completed the requirement analysis phase. Next, we shall do system design. Requirement analysis gives us the product specifications and its basic modules. System design takes these inputs to evolve the technical drawing of the product.

During system design, our main task is to explore various technical alternatives and finally select the basic technical configuration of the product. Each of the steps in the

ACTIVITY, DATA ACCURACY REQUIREMENTS AND DELIVERABLE OF EACH STAGE			
Stage	Main Activities	Data Accuracy	Deliverable
Requirement Analysis	Understand basic requirements from the product	Ball park figures for technical and financial analysis	- Product specification - List of modules - Rough business case
System Designing	Technology selection	Published figures from supplier catalogue and technical literature	- Product block diagram - Basic POC model
Module Design	Technical drawing	Published figures from supplier catalogue and technical literature	- Engineering drawing - Detailed business case
Prototype	- Develop prototype - Select suppliers - Make or buy decisions	Accurate technical details, fairly accurate market projection	- Bill of material - Manufacturing plan
Unit Testing	- Validate prototype in controlled environment - Estimate product price and market demand	Accurate technical details, accurate market projection	- List of suppliers - User manuals - Marketing plan
Integration Testing	- Performance testing - Distribution channel selection	Negotiated contracts with suppliers	- Detailed manufacturing plan - Quality assurance plan - Product distribution plan - Product packaging design
Field Testing	- Sample testing in real life use - Product launch planning - Stocking finished products at sales outlet	Negotiated contracts with suppliers and distributors	- After sales support plan - Manufacturing schedules - Sales force training material

Vee model has specific tasks and deliverables.

As we progress in these stages, we also make our cost and profitability projections more and more accurate. The initial stages start with rough ballpark estimates. Initially, we have little idea about the product, so we select our alternatives using published data and guidelines. As we get more and more clarity on our final product, we also collect accurate information. The table above gives the details of the activity, data accuracy requirements, and the deliverable of each stage.

Gateway review

In the Vee development model, we re-evaluate our business plan at the end of each stage. We evaluate it for technical and commercial viability after incorporating our new learning. The learning will come from the collected information and experience gained from the prototype development, trial, and the environment.

Product development is a time taking activity. Many things change

during the development of a project. New technologies emerge, policies shift, prices change, and other competitors may come out with some new products. All these changes have an impact on the business case of the product.

At the gateway review, a decision will be taken on Go, Rework, or Drop the development project. An assessment of the risks and an honest evaluation of the project are always required.

Delay in calling a nonviable product development project causes unnecessary investment and time. If we look around, there are plenty of examples of companies that have gotten into financial trouble for such delays in dropping their nonviable business plans. Periodic evaluation and risk assessment are equally important as technical development.

In our next article, we shall use these concepts for evaluating various technical options and learn about the economic evaluation of the product development project while developing the system design. **EFY**